Yeah, yeah, yeah, we We'd like to continue there's not much more to say this afternoon It's the first lecture and there's nothing to we're not that advanced that we can do exam questions But observation that mark shared with me, I think it's a good observation We love that you come to us questions in the break We really love it because it shows that you want to understand and maybe you don't understand and we do something here There we have a feeling mark said it to me now that From many of these questions if you ask them When everybody's listening then there are other people here Which maybe did not dare to ask the question and we'll get a lot from the questions now. I know it's exposure I realize it and I don't want to point that people say you asked me a good question there But it would be very valuable if you have these thoughts in your head and you say why why why why? You ask them planery and not personally, of course, you're always welcome to come It's not in any way rejecting your interest in your question But there's a lot of value in you coming with a question that probably other students have Which are not either they haven't realized they don't understand or they're not there Whatever we're not judging so consider asking questions in this format as well or Sometimes after and maybe if it's okay with you we can tell you after we explain to you here Can you ask it again after the break? So we can explain to other people and sometimes we remember we do it ourselves and sometimes we don't so please Please do that. Yeah, because what I'm a bit worried about is that we are not able We have 15 minutes of a break. Yes, but we have a lot of students that have but you really want to answer your questions But I'm a bit afraid that the 15 minutes are not sufficient to go through all your questions. So I really want to give it a space That and within this time we can use that space. So if their questions we can answer them now, that's easy. You know problem But 15 minutes, that's the limit of the break.

So that's yeah. Yeah Just so you know and there's a lot of value in the questions you ask That's what we also want to tell you that we see through your questions what we have failed to explain what we can explain better And that's that's how we improve so and if you just brain waves I don't know not always good with these kind of things But if you also say, okay, but I don't want to ask my question out loud. I don't want to be fine I propose something now and you give me feedback I'm I have no issue opening my laptop during the break and look at this court And if everybody has a question to the lecture that we just had and he doesn't want to approach it doing the instruction sessions But wants to set it in discord fine I'm happy to do that as well reading your comments and then we come back to the question. So then it's still Anominous while we can still discuss your questions.

That's something that would resonate with you or do you say no, that's that's just stupid Yes, some of some students feels good.

Okay, let's try we can try we are experimenting. Yes. Yeah. Yeah, okay Yeah, and and my apologies for all of you are better at math than me which found also the discrepancy in my minus signs The plus signs I apologize It's not because I don't appreciate math But I'm not very good at it.

I'm an engineer So the physicists here are Smiling and a bit disappointed, but I try to get the essence of it And if I miss the minus sign or plus sign also feel free to shout at me You miss something?

It's okay. I don't get offended at all So that's really important is it because I'm working with a very Limited kind of I work with this basically and I try to include in every topic the important things that are to be discussed So we almost finished. There's one more topic. Maybe it's not a topic at all It's the fact that in engineering and I think you've got this somewhere already in your curriculum We don't use what we like using DB's For describing the change the relationship between values Typical number will be DB for a gain of an amplifier.

We'll say the amplifiers 10 DB gain How much is 10 DB gain? If an amplifier is 10 DB gain how how much strong is the signal at the output compared to the input? 100 times one guess more guesses if it the signal at the output is 10 DB stronger Then it so that there's a 10 billion No Times 10 increase.

Yes, exactly. So 10 DB is 10 times more power Okay How do you know because there's a formula I didn't I don't invent this Formula and we will use it a lot in this course and in many many questions also in exams We will use DB To indicate all sorts of properties For example the gain of an antenna could be in DB right mark happened before Yes, doesn't yes so if P out divided by P in Yeah, this is the ratio and it has no units Just the number If I want to express this in DB, I will take 10 log base 10 of P out divided by P in Okay, so if the power at the output is double the power the input Double What will be the gain in DB? Take your calculator practice.

Take your calculator. Take you all have calculators or your laptop So whatever you're using and try to give me the number roughly.

What is if P out? Is two times P in P out in DB Is P in plus DB About 3 DB.

Thank you By the way, sorry that I interfere But the comment that I just made about take your calculator and who I it's it sounds stupid But it's really important that you check that you know how to do that with your calculator It happens every exam that we have students for the first time taking that calculator and try to calculate Logarithmic scale to the base of 10 and then they realize oh I cannot calculate it logarithmic to the base of 10 with my calculator Oh, I have no idea how to do that with my calculator. How do I calculate that? We will not answer these questions if you're not able to use the calculator yourself. That's a challenge So makes it yes simple calculators, so the basic ones the FX 82 Family or whatever the equivalent is for not for Casio FX 82 I've been working for the last 40 years.

It always works Yeah, okay, so that that's that's Easy and again the way to calculate you do on your calculator log 10 of 2 Which gives you 0.3 or 1 something and then you multiply this by 10 And you get roughly 3 DB okay, and if I ask you If I tell you that the p out in DB is equal to p in Plus 6 DB And I asked you to tell me what is the ratio between p out of p in? What is this value? How much is this Again use your calculators don't guess you also guess actually, but okay You can guess here actually guess here.

That's a nice part about it two times.

It's 3 DB What is 6 DB two times two? three plus three Four yes Actually engineers become very good at this actually for every DB single-digit DB.

There is a That you say brujeso. I don't know what the English you have a memory kind of thing that you memorize all of these doesn't matter So this is very useful yes, please sorry sorry yeah Yes, yes, thank you for saying asking this when you calculate the ratio between voltages Use 20 But this is ratio between powers Don't ask me why I didn't invent the rules I? Didn't really seriously. I didn't invent it. I didn't know now. There is a reason there is a deeper reason if you want If you go from voltages if you have if you have a power You know the power in electricity and electronics power Is proportional to the voltage squared, and that's why the 20 DB comes from That's exactly where it comes from because if you take the log 10 log of power in in power And you replace the power with voltage you get 20 log because the power of 2 goes out of the log Get 20 log the voltage ratio Okay, so that's where the 20 comes from and we call these units not DB. We call them DB volts Because they represent a voltage gain and not the power gain and these are not the same things, but I'm sorry. I hear I need to interfere because sometimes always something goes wrong with the DB and DB So I'm really sorry, but this is important because in the in the exams that also so that mentioned DBV so the DB volt DBV And we so far we discussed DB DB Which is a ratio? But it's not a power yet.

We are talking about pass But it's not a power yet when we in this course. We talked then about DB M's yes, and DB watts yes, and The only difference is a DB is just a ratio between it's a multiplication So it's oh, it's a factor of two. It's a factor of five. It's a factor of whatever That's a ratio on how much power that you gained if you what it makes it so easy if you work with the beast Then you get out of a multiplication, so it's three times the power you can just Add five words it and then you add it so then then it's a much easier calculation And you will see in this course. We will do a lot with DB. However This is unitless This has you this is a power you cannot convert DB to DB M.

I see that every time that students argued discuss or whatever Oh, I converted that to that no way possible.

You can convert DB M to DB watt, but we will make clear That there's always an add-on DB is a unitless Value we are not using that to express a power.

It's a ratio yeah Every year, and I don't know where that please if you know where that comes from every year someone says okay, but DB is so Plus 30 is equal to DBM I Have no idea where that comes from I have no idea who told you that that's wrong No way you cannot convert DB to DBM The same statement.

No it would work if it's DB watt, but it still doesn't work. It doesn't matter No, no because don't confuse them.

No because Yes, that's a different thing.

Yeah, that's a different thing I'm not DB and people confuse DB and DB watts that's yes because DBM is relative to 1 milliwatt of power so if you Actually, we do yeah, we don't yeah, so if you want if you have a power So this is these are all ratios all this this sketch is in ratios But if you want to know what the power is you can also use the same logarithmic scale again You might ask yourself. What's the purpose? Why do we do it? multiple reasons some of them have to do with fact that all the Components in an RF system will have their gains their their noise figures all in DB all the things are specting DB So if you have output power of 2 milliwatts, and you need to put in 10 of 10 DB afterwards How much power do you have very difficult, but if you know it's 3 DBM plus 10 DB, and it's 13 DBM You're somewhere so all this exchange of units is not done So we work predominantly in DBM in engineering and DBM is nothing more than 10 log 10 of the power divided by 1 milliwatt Milliwatt so 0 DBM is How much power is 0 DBM is very confusing always when I say 0 people think it's nothing. Yes It's 1 milliwatt and minus 10 DBM Suddenly there's minus power Why not? minus 10 DBM Is how much? sorry Yeah, 0.1 milliwatt most of the Signals you're detecting with your mobile phone are very low minus DB m's there are applications You can put on your phone to give you the power levels of your Wi-Fi the wireless network the 5G network Minus 70 DBM is a very typical number for this so very low power levels And we talk about the implication of low power the error correction and modulation all that stuff There is a quiz in Canvas you can do to practice DBM and DB There will be on Thursday morning when we meet we'll spend the first couple of minutes you good doing an entry-level DBM quiz, so you refresh DBM and Thursday we start handling sampling Right anything else we forgot to say no that that was for me quite important because I every year I have to subtract points or we have to subtract points from the exam due to these stupid mistakes that yes Students mix up DB and DBM.

It is worthwhile to spend some time on it And normally we assume that you know that so it's not that we spend a lot on time of explaining it But I really hope that this year we can make an improvement on that by giving a brief recap on it Yeah between now and Thursday There anything it's again.

It's in the module in Canvas you can practice which is good You can use the reader to refresh what you talked about and take all my math mistakes out of the of your notes Because there must be some of them The more you use as I said in the beginning the more you use the quizzes the better We know where you are as a group of students The better we can tailor what we provide here to what you need So even if you're not doing it for yourself because the oh I know this Take the extra 15 minutes 20 minutes These are not long quizzes usually most of you if you're very good at the call and the content you'll take maybe five minutes or ten Do this so we have a better picture of where you are students so we can really get you the right information I think that's it, and we will try to find something from an exam for next lesson I'm not sure we'll manage for something is a bit I I think actually they have once we finish the second lecture. We have one exam question that we can do all right so next Thursday we'll have an exam question in the lecture, so Yeah, anything else mark? No, the only thing is when you study at home, and you have the impression that there's something unclear or not Plus please let us know we are also here to provide the best Content for you guys, and if you really have the impression, okay? But something doesn't add up, and I don't know what to read I don't have to look at let us know Yeah Seems that then our information is not self-explaining and we really would like to give you the best opportunity to study for the course question Yes, yes, they're recording lectures of previous two years, and there will be recordings of this session as well and every session in the course Everything that happens in this room and in the other room should be recorded for the completeness Which means that today will finish at 10 to 4 and maybe and Thursday will finish at 11 30 I don't know depending on what what we do We fill the time as we need yes now you can these two you can convert because What you can express if you have one one thousand milliwatt of power It's one watt so that's to say that's eat That's just you can express that as well in DBM or in DB watt Yes, if if you just follow the calculation So if you put it in and you have for example one watt and you express it in DB watts That's a zero DB what that's indeed zero DB watts, and if you express the same number in DBM So if you just put it in calculation because the reference is then to one DBM, so it's to the power of 1,000 So log to the log to the scale of 10 to the power of 1,000 Then you actually end up with 30 DBM, and that's where I think some of that confusion comes from yeah Yeah, so I just lost my you put it down already.

Yeah, so this so zero DB what is Equal 30 DBM to 30 DB.

Yes, and this is fine because they're both powers But you cannot convert it in any way to DBM at a DB That's because that's just a ratio.

It's a unit unit less Expression it's a factor Yes, and some people confuse it yes, sometimes they use this and I don't know why but That's why I want to make that certain If you yeah, so if you find the source where it is mentioned differently, please let me know I really would like to point it out to the person who is giving that information last questions suggestions requests All right, then see you Thursday at quarter to night night Yeah